Matroid Intersection Polytope

$M_1 = (E, \mathcal{I}_1)$, $M_2 = (E, \mathcal{I}_2)$

$P(M_1 \cap M_2) = \operatorname{conv}\{\chi^I : I \in \mathcal{I}_1 \cap \mathcal{I}_2\}$

Thm: $P(M_1 \cap M_2) = P(M_1) \cap P(M_2)$

Recall: $P(M) = \{x \geq 0,\ x(F) \leq r(F),\ \forall F \subseteq E\}$

This theorem implies:

$$P(M_1 \cap M_2) = \{x \geq 0,\ x(F) \leq r_1(F),\ x(F) \leq r_2(F),\ \forall F \subseteq E\}.$$

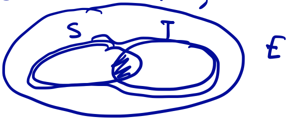
Tight Constraints

Defn: Let $x \in P(M)$.

$U \subseteq E$ is called tight if $x(U) = r(U)$.

Lemma 1: For $x \in P(M)$, if $S, T \subseteq E$ are tight,

then so are $S \cap T$, $S \cup T$



pf: $r(S \cup T) + r(S \cap T) \geq x(S \cup T) + x(S \cap T)$

$$= x(S) + x(T)$$

$$= r(S) + r(T)$$

$$\geq r(S \cup T) + r(S \cap T)$$

Must have equality throughout.

Chains: Nested Sets

Lemma 2: Let $x \in P(M)$, let $\mathcal{C} = \{C_1, \ldots, C_k\}$ an inclusion-wise maximal chain of tight sets:

chain: $\phi \subseteq C_1 \subseteq C_2 \subseteq \cdots \subseteq C_k$, tight $\Leftrightarrow$ $x(C_i) = r(C_i)$.

Then: any set $T \subseteq E$ that is tight must satisfy:

$$\chi^T \in \text{span}\{\chi^C : C \in \mathcal{C}\}$$

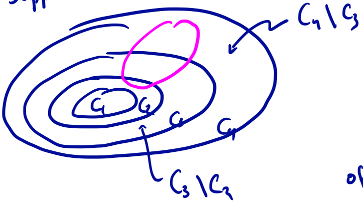
pf: Suppose not. Then $\exists T$ tight with $\chi^T \notin \text{span}\{\chi^C : C \in \mathcal{C}\}$.

If $T \not\subseteq C_k$, let $\hat{C} = C_k \cup T$. Then prev. lemma (Lemma 1) says $\hat{C}$ tight: $C_1 \subseteq \cdots \subseteq C_k \subseteq \hat{C}$.

Lemma 2 proof — cont —

Suppose $T \subseteq C_k$.

$$T \subseteq \bigcup C_{i+1} \setminus C_i$$



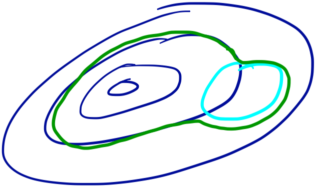
First: T is a union of several of these increment sets.

$$T = \bigcup_{j \in J} C_{j+1} \setminus C_j$$

Then: $$\chi^T = \sum_{j \in J} \chi^{C_{j+1} \setminus C_j} = \sum_{j \in J} \left(\chi^{C_{j+1}} - C^{C_j} \right) \in \text{span}\{\chi^C : C \in \mathcal{C}\}$$

Lemma 2 proof — cont —

Therefore: $\exists j$ with $T \cap (C_j \setminus C_{j-1}) \subseteq C_j \setminus C_{j-1}$



Again we can use Lemma 1 to contradict maximality of $\mathcal{C}$.

Let: $\hat{C} = (T \cap C_j) \cup C_{j-1}$

This is again tight, and $C_{j-1} \subsetneq \hat{C} \subsetneq C_j$.